

磁场量与静态、时变场

（承上页）自由空间中，一条沿z方向流过恒定电流I的无限长直导线产生的磁通量密度为：

$$\mathbf{B} = \hat{\phi} \frac{\mu_0 I}{2\pi r} \quad (\text{T}), \quad (1.13)$$

r为到导线的径向距离；带帽号的φ为方位角单位矢量，表示磁场沿环绕电流的圆周切向。磁通量密度的单位是特斯拉（T），得名于电气工程师Nikola Tesla（1856—1943）。他在变压器等方面的工作推动了低损耗远距离输电。μ₀为真空磁导率，原书取4π × 10⁻⁷ H/m，它与介电常数ε₀相对应。两者的乘积决定真空光速：

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \quad (\text{m/s}). \quad (1.14)$$

如第1-3.2节式（1.11）所述，电荷q'在外电场 $\mathbf{E}$ 中受到电场力 $\mathbf{F}_e = q'\mathbf{E}$ 。处于磁场 $\mathbf{B}$ 中的电荷只有在运动且速度 $\mathbf{u}$ 不与 $\mathbf{B}$ 平行或反平行时，才受到非零磁场力 $\mathbf{F}_m$ 。该力同时垂直于 $\mathbf{B}$ 和 $\mathbf{u}$ ，详见第5章。

在原书采用的均匀线性介质模型中，将式（1.13）的μ₀换为介质磁导率μ，可描述相应介质中的场。多数天然材料近似为非磁性材料，μ ≈ μ₀；铁、镍等铁磁材料的μ可远大于μ₀。μ表征材料的磁化响应，常写为：

$$\mu = \mu_r \mu_0 \quad (\text{H/m}), \quad (1.15)$$

μ_r为无量纲的相对磁导率，常用铁磁材料的值见附录B。第二对基本场量是 $\mathbf{B}$ 与磁场强度 $\mathbf{H}$ ，在此介质模型下满足：

$$\mathbf{B} = \mu \mathbf{H}. \quad (1.16)$$

1-3.4 静态场与时变场

电磁量是否以及如何随时间变化至关重要。为此约定以下术语：

静态：量不随时间变化。dc（直流）有时也用于描述不随时间变化的其他电磁量。

动态或时变：量随时间变化，但不限定变化的具体形式。

波形：量的数值随时间变化的曲线。

周期性：波形每隔周期T重复，例如正弦波和方波；满足通常傅里叶展开条件的周期波形可分解为正弦分量之和。

正弦变化：量随时间作正弦或余弦变化；本书以ac讨论此类激励。

电场与电荷q有关，磁场与电流I = dq/dt有关，两者是否耦合须区分静态和时变情形。先考虑恒定电流：取匀速带电粒子束的一小段，其中的总电荷量决定相应电场，而穿过截面的电荷流率（电流）决定相应磁场。少量电荷高速运动与大量电荷低速运动可以形成相同的电流。（接下页）

校注：交流并不必然是正弦波；此处限定为本书的正弦稳态讨论。光速3 × 10⁸ m/s为近似值。2019年SI重新定义后，μ₀的数值不再精确等于4π × 10⁻⁷ H/m，但该近似足以用于本书的常规计算。BIPM：https://www.bipm.org/en/history-si/ampere。

a very long wire residing in free space leads to the result that the **magnetic flux density** $\mathbf{B}$ induced by a constant current I flowing in the z direction is given by

$$\mathbf{B} = \hat{\phi} \frac{\mu_0 I}{2\pi r} \quad (\text{T}), \quad (1.13)$$

where r is the radial distance from the current and $\hat{\phi}$ is an azimuthal unit vector expressing the fact that the magnetic field direction is tangential to the circle surrounding the current. The magnetic field is measured in tesla (T), named in honor of Nikola Tesla (1856–1943). Tesla was an electrical engineer whose work on transformers made it possible to transport electricity over long wires without too much loss. The quantity μ_0 is called the **magnetic permeability of free space** [$\mu_0 = 4\pi \times 10^{-7}$ henry per meter (H/m)], and it is analogous to the electric permittivity ϵ_0 . In fact, as we will see in Chapter 2, the product of ϵ_0 and μ_0 specifies c , which is the **velocity of light in free space**:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \quad (\text{m/s}). \quad (1.14)$$

We noted in Section 1-3.2 that, when an electric charge q' is subjected to an electric field $\mathbf{E}$, it experiences an electric force $\mathbf{F}_e = q'\mathbf{E}$. Similarly, if a charge q' resides in the presence of a magnetic flux density $\mathbf{B}$, it experiences a **magnetic force** $\mathbf{F}_m$, but only if the charge is in motion and its velocity $\mathbf{u}$ is in a direction not parallel (or anti-parallel) to $\mathbf{B}$. In fact, as we learn in more detail in Chapter 5, $\mathbf{F}_m$ points in a direction perpendicular to both $\mathbf{B}$ and $\mathbf{u}$.

To extend Eq. (1.13) to a medium other than free space, μ_0 should be replaced with μ , which is the **magnetic permeability** of the material in which $\mathbf{B}$ is being observed. The majority of natural materials are **nonmagnetic**, meaning that they exhibit a magnetic permeability $\mu = \mu_0$. For ferromagnetic materials, such as iron and nickel, μ can be much larger than μ_0 . The magnetic permeability μ accounts for **magnetization** properties of a material. In analogy with Eq. (1.11), μ of a particular material can be defined as

$$\mu = \mu_r \mu_0 \quad (\text{H/m}), \quad (1.15)$$

where μ_r is a dimensionless quantity called the **relative magnetic permeability** of the material. The values of μ_r for commonly used ferromagnetic materials are given in Appendix B.

► We stated earlier that $\mathbf{E}$ and $\mathbf{D}$ constitute one of two pairs of electromagnetic field quantities. The second pair is $\mathbf{B}$ and the **magnetic field intensity** $\mathbf{H}$, which are related to each other through μ :

$$\mathbf{B} = \mu \mathbf{H}. \quad (1.16)$$

1-3.4 Static and Dynamic Fields

In EM, the time variable t , or more precisely if and how electric and magnetic quantities vary with time, is of crucial importance. Before we elaborate further on the significance of this statement, it will prove useful to define the following time-related adjectives unambiguously:

- **Static**—describes a quantity that does not change with time. The term **dc** (i.e., direct current) is often used as a synonym for static to describe not only currents, but other electromagnetic quantities as well.

- **Dynamic**—refers to a quantity that does vary with time, but conveys no specific information about the character of the variation.

- **Waveform**—refers to a plot of the magnitude profile of a quantity as a function of time.

- **Periodic**—a quantity is periodic if its waveform repeats itself at a regular interval, namely its period T . Examples include the sinusoid and the square wave. By application of the Fourier series analysis technique, any periodic waveform can be expressed as the sum of an infinite series of sinusoids.

- **Sinusoidal**—also called **ac** (i.e., alternating current), describes a quantity that varies sinusoidally (or cosinusoidally) with time.

In view of these terms, let us now examine the relationship between the electric field $\mathbf{E}$ and the magnetic flux density $\mathbf{B}$. Because $\mathbf{E}$ is governed by the charge q and $\mathbf{B}$ is governed by $I = dq/dt$, one might expect that $\mathbf{E}$ and $\mathbf{B}$ must be somehow related to each other. As we learn later, they may or may not be interrelated, depending on whether I is static or dynamic.

Let us start by examining the dc case in which I remains constant with time. Consider a small section of a beam of charged particles, all moving at a constant velocity. The moving charges constitute a dc current. The electric field due to that section of the beam is determined by the total charge q contained in it. The magnetic field does not depend on q , but rather on the rate of charge (current) flowing through that section. Few charges moving very fast can constitute the same

Table 1-3 The three branches of electromagnetics.

Branch	Condition	Field Quantities (Units)
Electrostatics	Stationary charges ($\partial q/\partial t = 0$)	Electric field intensity E (V/m) Electric flux density D (C/m ²) D = ϵ E
Magnetostatics	Steady currents ($\partial I/\partial t = 0$)	Magnetic flux density B (T) Magnetic field intensity H (A/m) B = μ H
Dynamics (time-varying fields)	Time-varying currents ($\partial I/\partial t \neq 0$)	E, D, B, and H (E, D) coupled to (B, H)

current as many charges moving slowly. In these two cases, the induced magnetic field is the same because the current I is the same, but the induced electric field is quite different because the numbers of charges are not the same.

Electrostatics and *magnetostatics* refer to the study of EM under the specific, respective conditions of stationary charges and dc currents. They represent two *independent* branches, so characterized because the induced electric and magnetic fields do not couple to each other. *Dynamics*, the third and more general branch of electromagnetics, involves *time-varying fields* induced by time-varying sources, that is, currents and associated charge densities. If the current associated with the beam of moving charged particles varies with time, then the amount of charge present in a given section of the beam also varies with time, and vice versa. As we will see in Chapter 6, the electric and magnetic fields become coupled to each other in that case.

► A time-varying electric field generates a time-varying magnetic field, and vice versa. ◄

Table 1-3 provides a summary of the three branches of electromagnetics.

The electric and magnetic properties of materials are characterized by the parameters ϵ and μ , respectively. A third fundamental parameter also needed is the *conductivity* of a material σ , which is measured in siemens per meter (S/m). The conductivity characterizes the ease with which charges (electrons) can move freely in a material. If $\sigma = 0$, the charges do not move more than atomic distances, and the material is said to be a *perfect dielectric*. Conversely, if $\sigma = \infty$, the charges can move very freely throughout the material, which is then called a *perfect conductor*.

Table 1-4 Constitutive parameters of materials.

Parameter	Units	Free-Space Value
Electrical permittivity ϵ	F/m	$\epsilon_0 = 8.854 \times 10^{-12}$ $\approx \frac{1}{36\pi} \times 10^{-9}$
Magnetic permeability μ	H/m	$\mu_0 = 4\pi \times 10^{-7}$
Conductivity σ	S/m	0

► The parameters ϵ , μ , and σ are often referred to as the *constitutive parameters* of a material (**Table 1-4**). A medium is said to be *homogeneous* if its constitutive parameters are constant throughout the medium. ◄

Concept Question 1-1: What are the four fundamental forces of nature, and what are their relative strengths?

Concept Question 1-2: What is Coulomb’s law? State its properties.

Concept Question 1-3: What are the two important properties of electric charge?

(continued on p. 22)

电磁学分支与介质本构参数

(承上页) 这两种粒子束的电流*I*相同，因而产生的磁场相同；但其电荷数量不同，电场就不同。静电学研究静止电荷，静磁学研究恒定电流。在这些条件下，相应的电场与磁场不通过时变项相互耦合，因而可作为两个独立分支研究。

更一般的电动力学研究时变源（电流及有关电荷密度）所产生的时变场。原书以带电粒子束为例讨论电荷量与电流的时间变化，并指出此时须考虑电场与磁场的耦合，详见第6章。

表1-3 电磁学的三个分支（按原书概括）

分支	源的条件	场量及关系
静电学	静止电荷; $\partial q/\partial t = 0$	电场强度 E (V/m) 电通量密度 D (C/m ²) D = ϵ E
静磁学	恒定电流; $\partial I/\partial t = 0$	磁通量密度 B (T) 磁场强度 H (A/m) B = μ H
电动力学（时变场）	时变电流; $\partial I/\partial t \neq 0$	E, D, B, H 电场与磁场相互耦合

适用条件校注：上表是原书的入门分类，不是一般场问题的充分判据。由连续性方程，某体积内电荷的变化率取决于穿过其边界的净电流，而非某截面电流是否随时间变化。时变场的耦合应以麦克斯韦方程中的时间导数项理解，不能把“时变电场必然产生时变磁场，反之亦然”作为无条件命题。

介质的电学、磁学性质分别由 ϵ 、 μ 表征，还需要电导率 σ ，单位为西门子每米（S/m）。电导率描述电荷在介质中形成传导电流的能力。 $\sigma = 0$ 对应理想电介质：电荷不能产生宏观传导，但仍可出现原子尺度的极化位移。 $\sigma \rightarrow \infty$ 是理想导体模型。

表1-4 介质的本构参数

参数	单位	真空值（原书近似）
介电常数 ϵ	F/m	$\epsilon_0 \approx 8.854 \times 10^{-12}$ $\approx 10^{-9}/(36\pi)$
磁导率 μ	H/m	$\mu_0 \approx 4\pi \times 10^{-7}$
电导率 σ	S/m	0

ϵ 、 μ 和 σ 称为介质的本构参数。若本构参数在介质各处取值相同，则称该介质为均匀介质。

概念题

1-1: 自然界有哪四种基本相互作用？原书给出的相对强度分别是多少？

1-2: 什么是库仑定律？说明其性质。

1-3: 电荷的两项重要性质是什么？

本节概念题续于原书第22页。

Technology Brief 1: LED Lighting

After lighting our homes, buildings, and streets for over 100 years, the incandescent light bulb created by Thomas Edison (1879) will soon become a relic of the past. Many countries have taken steps to phase it out and replace it with a much more energy-efficient alternative: the **light-emitting diode (LED)**.

Light Sources

The three dominant sources of electric light are the incandescent, fluorescent, and LED light bulbs (Fig. TF1-1). We examine each briefly.

Incandescent Light Bulb

► **Incandescence** is the emission of light from a hot object due to its temperature. ◀

By passing electric current through a thin tungsten filament, which basically is a resistor, the filament's temperature rises to a very high level, causing the filament to glow and emit visible light. The intensity and shape of the emitted spectrum depends on the filament's temperature. A typical example is shown by the green curve in Fig. TF1-2. The tungsten spectrum is similar in shape to that of sunlight (yellow curve in Fig. TF1-2), particularly in the blue and green parts of the spectrum (400–550 nm). Despite the relatively strong (compared with sunlight) yellow light emitted by incandescent sources, the quasi-white light they produce has a quality that the human eye finds rather comfortable.

► The incandescent light bulb is significantly less expensive to manufacture than the fluorescent and LED light bulbs, but it is far inferior with regard to energy efficacy and operational lifetime (Fig. TF1-7). ◀

Of the energy supplied to an incandescent light bulb, only about 2% is converted into light, with the remainder wasted as heat! In fact, the incandescent light bulb is the weakest link in the overall conversion sequence from coal to light (Fig. TF1-3).

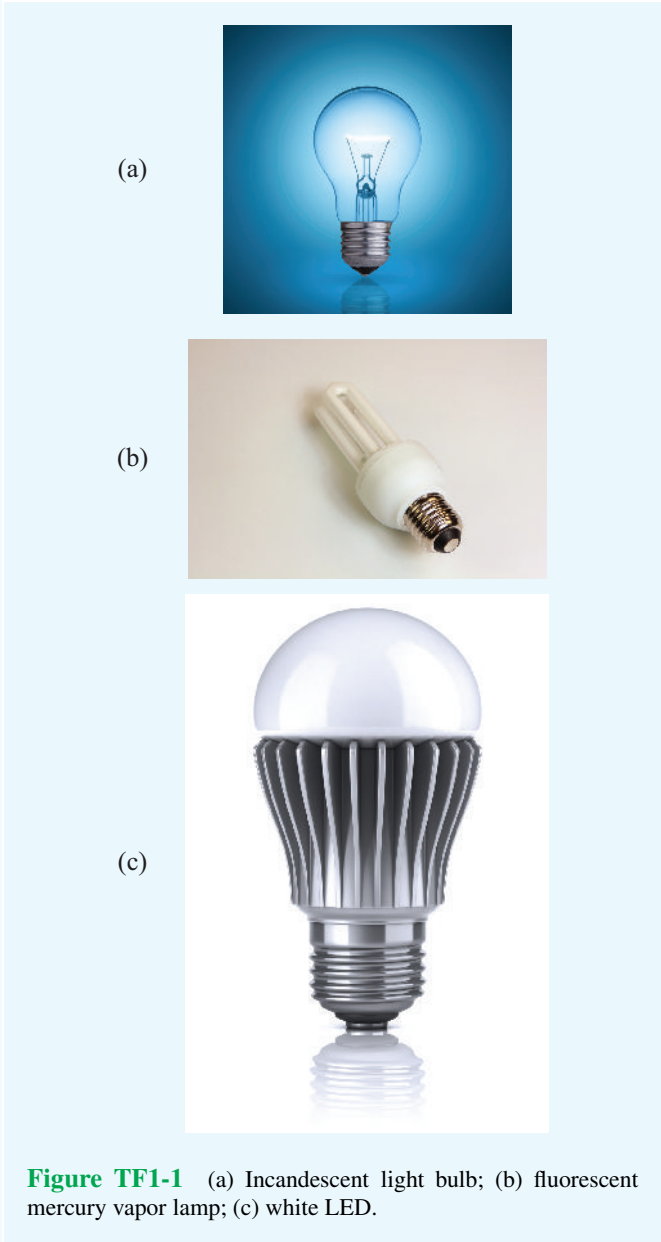


Figure TF1-1 (a) Incandescent light bulb; (b) fluorescent mercury vapor lamp; (c) white LED.

Fluorescent Light Bulb

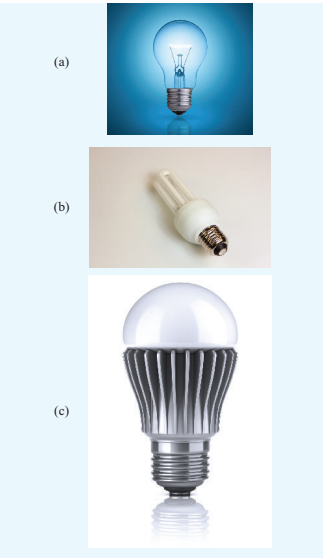
To **fluoresce** means to emit radiation in consequence to incident radiation of a shorter wavelength. By passing a stream of electrons between two electrodes at the ends of a tube (Fig. TF1-1(b)) containing mercury gas (or the noble gases neon, argon, and xenon) at very low pressure, the electrons collide with the mercury atoms, causing them to excite their own electrons to higher energy

技术简介1：LED照明

原书写道：自Thomas Edison于1879年研制实用白炽灯以来，白炽灯已照亮住宅、建筑物和街道一百多年，如今将逐步退出历史舞台。许多国家已采取措施，用更加节能的发光二极管（LED）替代白炽灯。

光源

本节比较三种主要电光源：白炽灯、荧光灯和LED灯，如图TF1-1。



图TF1-1 (a) 白炽灯；(b) 荧光汞蒸气灯；(c) 白光LED。

白炽灯

白炽发光是物体因温度很高而发出可见光的现象。电流通过细钨丝（本质上为电阻元件）时，灯丝温度升高，从而发光。发射光谱的强度和形状取决于灯丝温度。图TF1-2的绿色曲线给出一个典型例子；与太阳光的黄色曲线相比，两者在蓝、绿光区域（400—550 nm）的谱形有相似之处。尽管白炽灯的黄光成分相对更强，其近似白光通常给人舒适的视觉感受。

白炽灯的制造成本低于荧光灯和LED灯，但光效与使用寿命均明显较低，见图TF1-7。原书估计，输入白炽灯的能量仅约2%转化为可见光，其余主要以热的形式散失。因此，在从煤的化学能到可见光的转换链中，白炽灯是效率最低的环节，见图TF1-3。

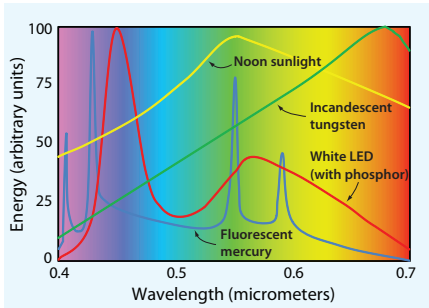
荧光灯

荧光是物质受到较短波长辐射激发后发射辐射的现象。荧光灯管中装有低压汞蒸气及有关工作气体。电子在灯管两端电极之间运动，并与汞原子碰撞，使原子的电子跃迁到较高能级。（接下页）

编校说明：本技术简介中的价格、寿命、光效及对2025年的预测均属于原书写作时的资料，不能视作当前产品参数。原书括号另列氖、氩、氙等气体；不同放电光源的工作气体与发光机制须分别理解。

LED照明：荧光转换与电致发光

(承上页) 受激电子回到基态时，会发射特定波长的光子，主要位于紫外波段。因此，汞灯光谱表现为若干窄谱线，见图TF1-2的蓝色曲线。



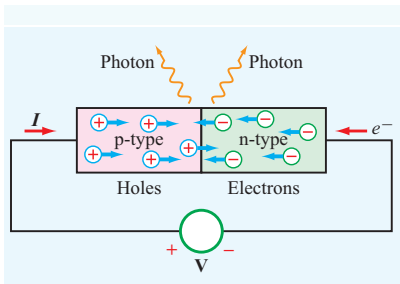
图TF1-2 常见可见光源的光谱。

图轴：横轴为波长 (μm)，纵轴为能量 (任意单位)。曲线：正午太阳光、白炽钨丝灯、荧光灯，以及使用荧光粉的白光LED。任意单位用于比较谱形，不代表绝对辐射功率。

为把汞的谱线转换为更接近白光的宽光谱，灯管内表面涂覆荧光粉。荧光粉吸收紫外辐射，再发射较长波长、分布较宽的光，从蓝光延伸至红光，这就是“荧光灯”名称的由来。原书此处举出掺钕钇铝石榴石 (YAG:Ce) 作为荧光材料的例子。

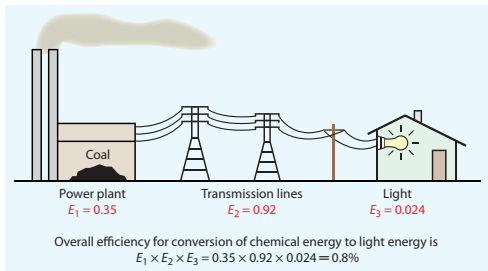
发光二极管

图TF1-1 (c) 中聚合物封装内的LED是在半导体芯片上形成的p-n结二极管。当结两端施加正向偏压时，电流通过结区；部分电子与空穴复合，并以光子的形式释放能量，见图TF1-4。



图TF1-4 电子与空穴的辐射复合产生光子。

图中Electrons: 电子; Holes: 空穴; Photon: 光子; p-type、n-type: p型、n型。校注：并非每一次电子—空穴复合都发光，还存在非辐射复合；这里描述的是LED中的辐射复合机制。



图TF1-3 从煤的化学能到可见光的转换效率。

图TF1-3中，煤经发电厂、电力传输和灯泡转换为光；各环节效率分别为 $E_1 = 0.35$ 、 $E_2 = 0.92$ 、 $E_3 = 0.024$ 。总效率为 $0.35 \times 0.92 \times 0.024 = 0.007728 \approx 0.8\%$ 。资料来源：美国国家研究委员会，2009年。

1-3 THE NATURE OF ELECTROMAGNETISM

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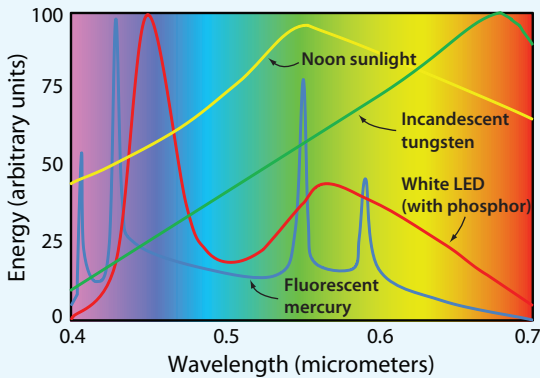


Figure TF1-2 Spectra of common sources of visible light.

levels. When the excited electrons return to the ground state, they emit photons at specific wavelengths, mostly in the ultraviolet part of the spectrum. Consequently, the spectrum of a mercury lamp is concentrated into narrow lines, as shown by the blue curve in Fig. TF1-2.

► To broaden the mercury spectrum into one that resembles that of white light, the inside surface of the fluorescent light tube is coated with phosphor particles [such as yttrium aluminum garnet (YAG) doped with cerium]. The particles absorb the UV energy and then reradiate it as a broad spectrum extending from blue to red; hence the name **fluorescent**. ◀

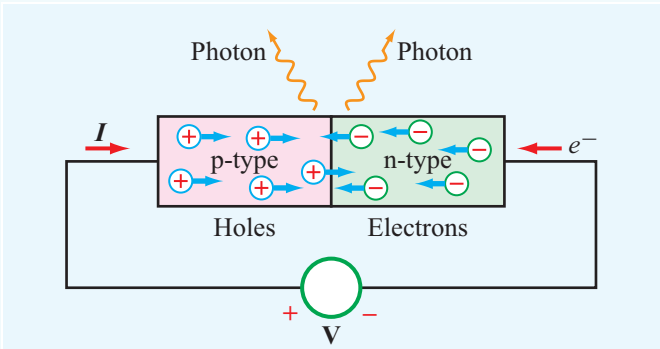


Figure TF1-4 Photons are emitted when electrons combine with holes.

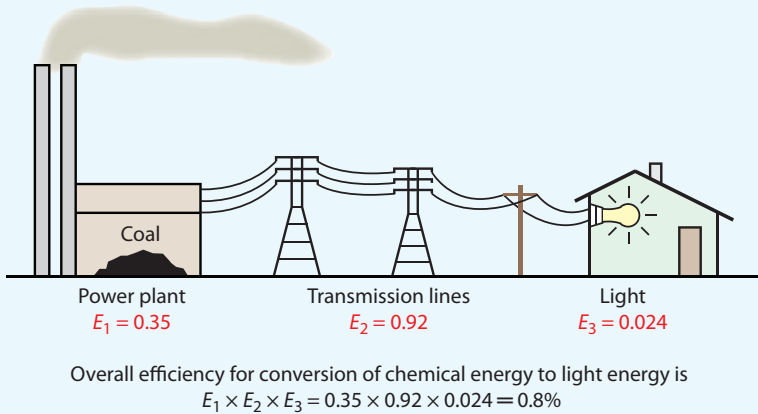


Figure TF1-3 Lighting efficiency. (Source: National Research Council, 2009.)

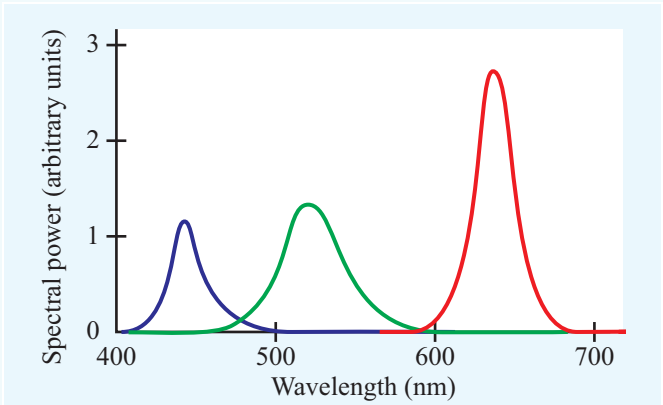


Figure TF1-5 The addition of spectra from three monochromatic LEDs.

► The wavelength of the emitted photon depends on the diode's semiconductor material. The materials most commonly used are aluminum gallium arsenide (AlGaAs) to generate **red light**, indium gallium nitride (InGaN) to generate **blue light**, and aluminum gallium phosphide (AlGaP) to generate **green light**. In each case, the emitted energy is confined to a narrow spectral band. ◀

Two basic techniques are available for generating white light with LEDs: (a) RGB and (b) blue/conversion. The RGB approach involves the use of three monochromatic LEDs whose primary colors (red, green, and blue) are mixed to generate an approximation of a white-light spectrum. An example is shown in **Fig. TF1-5**. The advantage of this approach is that the relative intensities of the three LEDs can be controlled independently, thereby making it possible to “tune” the shape of the overall spectrum to generate an esthetically pleasing color of “white.” The major shortcoming of the RGB technique is cost: manufacturing three LEDs instead of just one.

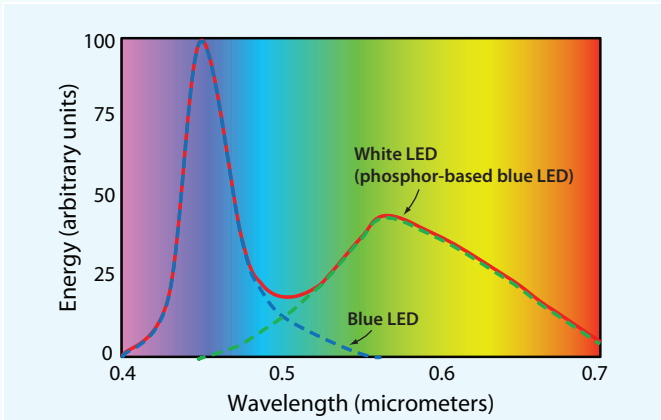


Figure TF1-6 Phosphor-based white LED emission spectrum.

With the blue LED/phosphor conversion technique, a blue LED is used with phosphor powder particles suspended in the epoxy resin that encapsulates it. The blue light emitted by the LED is absorbed by the phosphor particles and then reemitted as a broad spectrum (**Fig. TF1-6**). To generate high-intensity light, several LEDs are clustered into a single enclosure.

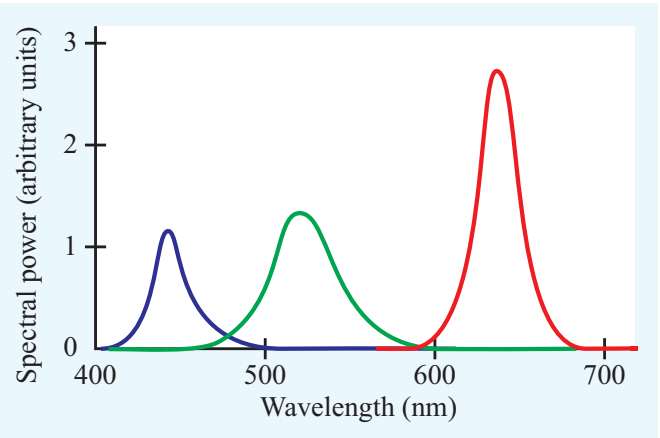
Comparison

► **Luminous efficacy** (**LE**) is a measure of how much light in lumens is produced by a light source for each watt of electricity consumed by it. ◀

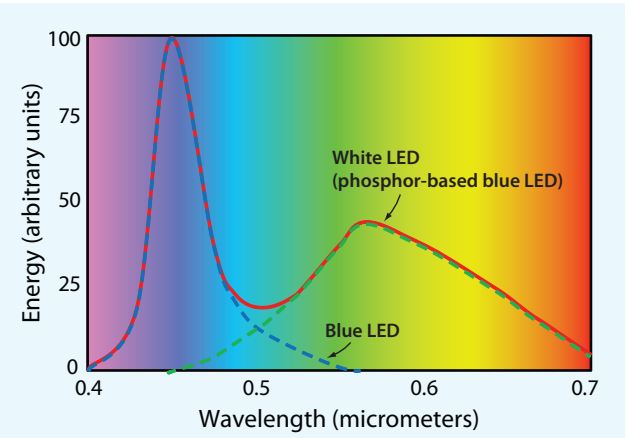
Of the three types of light bulbs we discussed, the incandescent light bulb is by far the most inefficient and its useful lifespan is the shortest (**Fig. TF1-7**). For a typical household scenario, the 10-year cost—including electricity and replacement cost—is several times smaller for the LED than for the alternatives.

白光LED与光效

LED发射光子的波长取决于半导体材料。原书列举：砷化铝镓（AlGaAs）用于产生红光，氮化铟镓（InGaN）用于蓝光，磷化铝镓（AlGaP）用于绿光。每一种单色LED的发射能量主要集中于较窄波段。



图TF1-5 三个单色LED的光谱相加。



图TF1-6 荧光粉转换型白光LED的光谱。

图TF1-5：横轴波长（nm），纵轴光谱功率（任意单位）。图TF1-6：横轴波长（ μm ），纵轴能量（任意单位）；曲线分别为蓝光LED与荧光粉转换型白光LED。化学式Al中的第二个字母是小写l（铝），不是大写I。

用LED产生白光有两种基本方法：（a）红、绿、蓝三色混光（RGB）；（b）蓝光激发荧光粉转换。RGB方法将三个单色LED的红、绿、蓝光混合，形成视觉上近似白光的输出，见图TF1-5。其优点是三色相对强度可以独立控制，从而调整总体光谱和白光色调。主要缺点是需制造、驱动三个LED，成本高于单LED方案。

蓝光LED / 荧光粉转换法把荧光粉颗粒分散在封装LED的环氧树脂中。部分蓝光被荧光粉吸收，再以较长波长的宽谱光重新发射，与剩余蓝光共同形成白光，见图TF1-6。要获得较高光输出，可以将多个LED集成于同一封装或灯具内。

比较

光效 (luminous efficacy, LE) 是光源输出的光通量与所消耗电功率之比，单位为流明每瓦（lm/W）。瓦特是功率单位，不能译成“每瓦电量”。

在原书比较的三类灯中，白炽灯光效最低、寿命最短，详见图TF1-7。按原书采用的家庭使用情景估算，将电费与更换费用合计，LED的10年使用成本仅为其他方案的一部分。

术语校注：光效（lm/W）与能量转换效率（无量纲）是不同物理量。RGB混光产生白色视觉感受，并不意味着其三峰光谱等同于太阳光的连续光谱。

Parameter	Type of Light Bulb			
	Incandescent	Fluorescent	White LED	
			Circa 2010	Circa 2025
Luminous Efficacy (lumens/W)	~12	~40	~70	~150
Useful Lifetime (hours)	~1000	~20,000	~60,000	~100,000
Purchase Price	~\$1.50	~\$5	~\$10	~\$5
Estimated Cost over 10 Years	~\$410	~\$110	~\$100	~\$40

Figure TF1-7 Even though the initial purchase price of a white LED is several times greater than that of the incandescent light bulb, the total 10-year cost of using the LED is only one-fourth of the incandescent’s (in 2010) and is expected to decrease to one-tenth by 2025.

LED照明：原书性能与成本比较

图TF1-7 灯具性能与10年使用成本

参数	白炽灯	荧光灯	白光LED 约2010年	白光LED 原书对约2025年的预测
光效 (lm/W)	约12	约40	约70	约150
使用寿命 (小时)	约1,000	约20,000	约60,000	约100,000
购买价格 (美元)	约1.50	约5	约10	约5
10年估算成本 (美元)	约410	约110	约100	约40

尽管白光LED的购买价格比白炽灯高数倍，原书估计其10年总使用成本在2010年前后仅为白炽灯的约四分之一，并预测到2025年前后降至约十分之一。

资料年代说明：表中“约2025年”是原书写作时的预测，不是对2025年实际市场的测量结果。本页只校对、翻译原书数据，不将历史预测改写成已实现的事实。

Concept Question 1-4: What do the electrical permittivity and magnetic permeability of a material account for?

Concept Question 1-5: What are the three branches and associated conditions of electromagnetics?

Exercise 1-1: Given charges $q_1 = 10\text{ mC}$, $q_2 = -10\text{ mC}$, and $q_3 = 5\text{ mC}$, all in free space, what is the direction of the force acting on charge q_3 ?

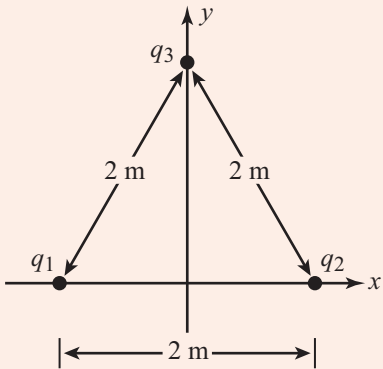


Figure E1.1

Answer: Along $+\hat{x}$ direction. [See  (the “” symbol refers to the book website: em8e.eecs.umich.edu).]

Exercise 1-2: Two parallel, very long, wires carry currents I_1 and I_2 . The magnetic field due to current I_1 alone is $\mathbf{B}_1$. What is the magnetic field due to both currents at a point midway between the wires if

- (a) $I_1 = I_2$ and both currents flow along the $+\hat{y}$ direction?
(b) $I_1 = I_2$, but I_2 flows along the $-\hat{y}$ direction?

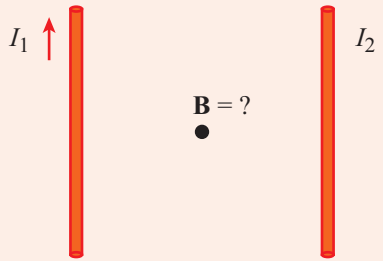


Figure E1.2

Answer: (a) $\mathbf{B} = 0$, (b) $\mathbf{B} = 2\mathbf{B}_1$. (See .)

1-4 Traveling Waves

Waves are a natural consequence of many physical processes: Waves manifest themselves as ripples on the surfaces of oceans and lakes; sound waves constitute pressure disturbances that travel through air; mechanical waves modulate stretched strings; and electromagnetic waves carry electric and magnetic fields through free space and material media as microwaves, light, and X-rays. All of these various types of waves exhibit a number of common properties, including:

- **Moving waves carry energy.**
- **Waves have velocity;** it takes time for a wave to travel from one point to another. Electromagnetic waves in a vacuum travel at a speed of $3 \times 10^8\text{ m/s}$, and sound waves in air travel at a speed approximately a million times slower, specifically 330 m/s . Sound waves cannot travel in a vacuum.
- **Many waves exhibit a property called linearity.** Waves that do not affect the passage of other waves are called **linear** because they can pass right through each other. The total of two linear waves is simply the sum of the two waves as they would exist separately. Electromagnetic waves are linear, as are sound waves. When two people speak to one another, the sound waves they generate do not interact with one another but simply pass through each other. Water waves are approximately linear; the expanding circles of ripples caused by two pebbles thrown into two locations on a lake’s surface do not affect each other. Although the interaction of the two circles may exhibit a complicated pattern, it is simply the linear superposition of two independent expanding circles.

Waves are of two types: **transient waves** caused by sudden disturbances and **continuous periodic waves** generated by a repetitive source. We encounter both types of waves in this book, but most of our discussion deals with the propagation of continuous waves that vary sinusoidally with time.

An essential feature of a propagating wave is that it is a self-sustaining disturbance of the medium through which it travels. If this disturbance varies as a function of one space variable, such as the vertical displacement of the string shown in **Fig. 1-10**, we call the wave **one-dimensional**. The vertical displacement varies with time and with the location along the length of the string. Even though the string rises up into a second dimension, the wave is only one-dimensional

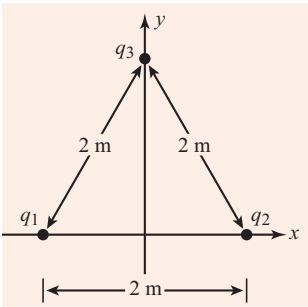
概念题、练习与1-4行波

概念题1-4: 介质的介电常数与磁导率分别表征什么性质？

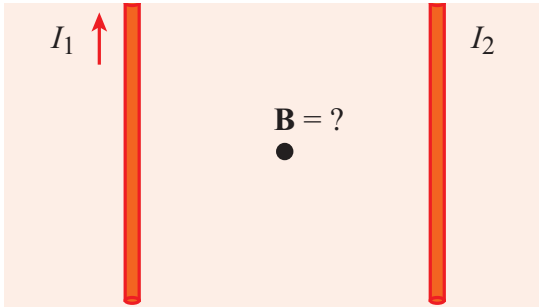
概念题1-5: 电磁学的三个分支及各自对应的条件是什么？

练习1-1: 自由空间中三个电荷的位置见图E1.1， $q_1 = 10\text{ mC}$ 、 $q_2 = -10\text{ mC}$ 、 $q_3 = 5\text{ mC}$ 。 q_3 所受合力指向何方？**答案：**沿 $+x$ 方向。

练习1-2: 两条平行、足够长的直导线分别通有电流 I_1 和 I_2 。在两导线中间的观察点，仅 I_1 产生的磁场为 $\mathbf{B}_1$ 。若（a） $I_1 = I_2$ ，两者均沿 $+y$ 方向；（b） $I_1 = I_2$ ，但 I_2 沿 $-y$ 方向，求总磁场。**答案：**（a） $\mathbf{B} = 0$ ；（b） $\mathbf{B} = 2\mathbf{B}_1$ 。



图E1.1 电荷的位置及间距。



图E1.2 平行载流导线。

原书EM符号指配套网站em8e.eecs.umich.edu。练习1-1的答案由库仑力的对称性核对；练习1-2由右手定则与磁场叠加核对。

1-4 行波

许多物理过程都会产生波：海洋、湖面的涟漪是水波；声波是在空气中传播的压力扰动；绷紧的弦可传播机械波；微波、光和X射线则是携带电场与磁场、能够在自由空间和物质介质中传播的电磁波。这些波具有若干共同性质：

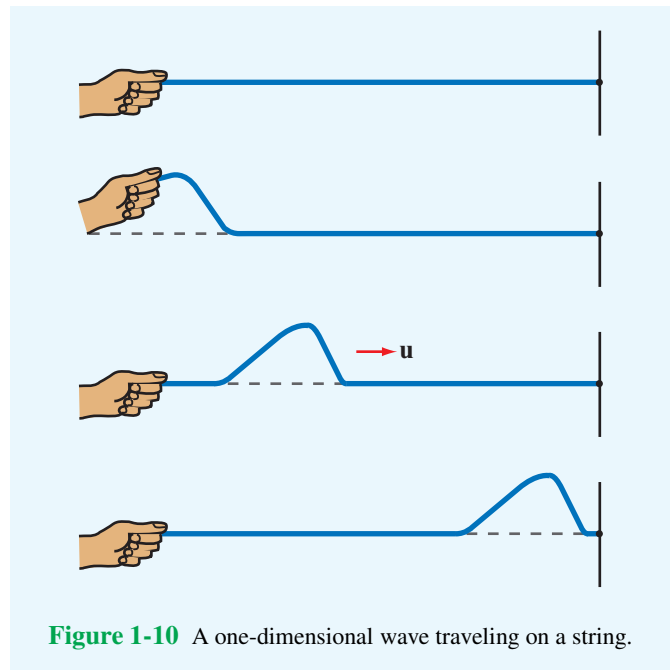
能量传输：行波携带能量。

有限传播速度：波由一点到另一点需要时间。真空中的电磁波速度约为 $3 \times 10^8\text{ m/s}$ ；原书取空气声速约 330 m/s ，比光速低约六个数量级。声波不能在真空中传播。

线性叠加：在线性条件下，两列波可相互穿过，总扰动等于两列波分别存在时扰动之和。日常对话中的声波、两个小石子激起的小振幅水面涟漪，可近似如此处理；干涉图样复杂并不意味着两列波的传播规律被改变。

波可分为突然扰动产生的瞬态波，以及周期性源产生的连续周期波。本书涉及两者，但主要讨论随时间作正弦变化的连续波。传播中的波是能够延续传播的扰动。若扰动只依赖一个空间坐标，即为一维波。例如弦的横向位移虽画在二维平面中，其数值仅随沿弦坐标及时间变化，因此仍是一维波。（接下页）

适用条件校注：线性叠加以介质及波动方程可作线性描述为前提；强声波或非线性介质中的电磁波不能无条件套用。电磁波在真空中传播不需要物质介质。



they include **plane waves**, **cylindrical waves**, and **spherical waves**. A plane wave is characterized by a disturbance that at a given point in time has uniform properties across an infinite plane perpendicular to its direction of propagation (**Fig. 1-11(b)**). Similarly, for cylindrical and spherical waves, the disturbances are uniform across cylindrical and spherical surfaces (**Figs. 1-11(b) and (c)**).

In the material that follows, we examine some of the basic properties of waves by developing mathematical formulations that describe their functional dependence on time and space variables. To keep the presentation simple, we limit our discussion to sinusoidally varying waves whose disturbances are functions of only one space variable, and we defer the discussion of more complicated waves to later chapters.

1-4.1 Sinusoidal Waves in a Lossless Medium

Regardless of the mechanism responsible for generating them, all linear waves can be described mathematically in common terms.

► A medium is said to be **lossless** if it does not attenuate the amplitude of the wave traveling within it or on its surface. ◀

because the disturbance varies with only one space variable. A **two-dimensional wave** propagates out across a surface, like the ripples on a pond (**Fig. 1-11(a)**), and its disturbance can be described by two space variables. By extension, a **three-dimensional wave** propagates through a volume, and its disturbance may be a function of all three space variables. Three-dimensional waves may take on many different shapes;

By way of an example, let us consider a wave traveling on a lake's surface, and let us assume for the time being that frictional forces can be ignored, thereby allowing a wave generated on the water's surface to travel indefinitely with no loss in energy. If y denotes the height of the water's surface

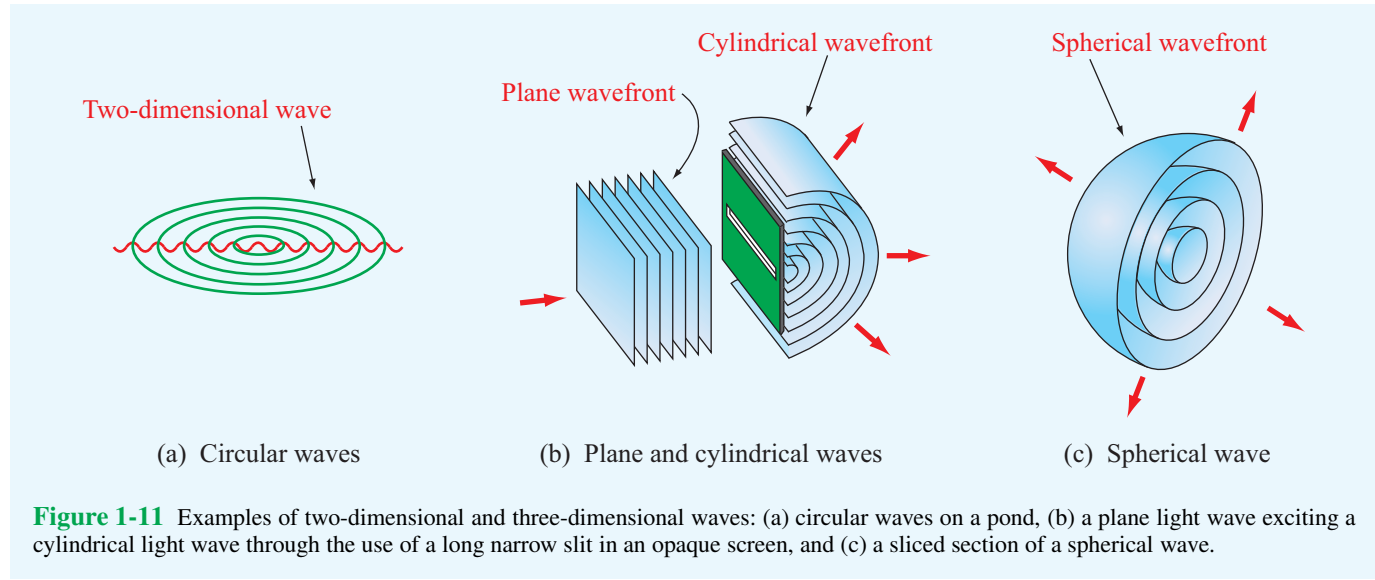


Figure 1-11 Examples of two-dimensional and three-dimensional waves: (a) circular waves on a pond, (b) a plane light wave exciting a cylindrical light wave through the use of a long narrow slit in an opaque screen, and (c) a sliced section of a spherical wave.

波的维数与无损介质中的正弦波

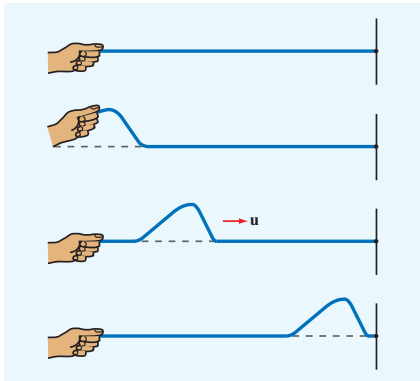


图1-10 沿弦传播的一维波， u 为传播速度。

(承上页) 一维波的扰动仅依赖一个空间坐标。二维波在表面上传播，例如池塘涟漪，需两个空间坐标描述。三维波在空间体积中传播，一般可依赖三个空间坐标，其形式包括平面波、柱面波和球面波。

在某一时刻，理想平面波在垂直于传播方向的整个平面上具有相同的场分布。原书同样用柱面、球面上的相同扰动说明理想柱面波和球面波，见图1-11。以下将用数学表达式描述波对时间与空间的依赖，先限于随时间作正弦变化、并仅依赖一个空间变量的情形；更复杂的波留到后续章节。

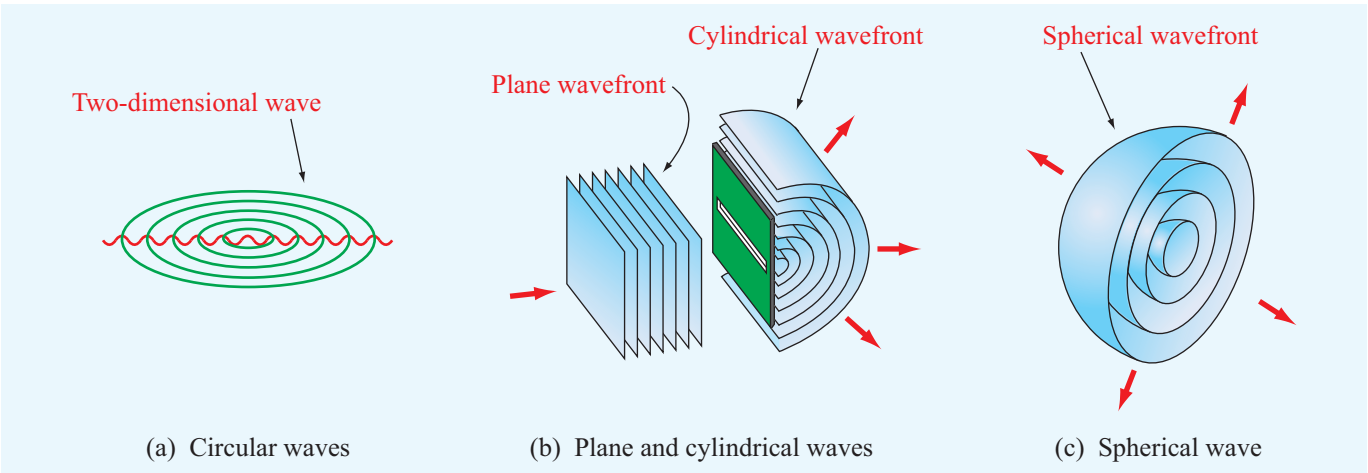


图1-11 (a) 池塘中的圆形波；(b) 平面光波穿过不透明屏上的狭缝，形成柱面光波；(c) 球面波的剖切示意。

图中Two-dimensional wave：二维波；Plane、Cylindrical、Spherical wavefront分别为平面、柱面、球面波前。波前形状与场矢量的方向是不同概念。

1-4.1 无损介质中的正弦波

无论产生机制如何，线性波都可用共同的数学形式描述。原书把无损介质描述为不使其中或其表面传播的波发生振幅衰减的介质。以湖面波为例，暂时忽略摩擦，便可研究不因耗散而损失能量的波。设 y 表示水面高度。(接下页)

定义校注：无损指没有介质耗散造成的能量损失。即使介质无损，球面波或柱面波仍可因几何扩散而随距离减小振幅；原书“不衰减振幅”的说法须限于排除了这类几何因素的情形。

正弦行波的相位与相速度

（承上页）设 y 为水面相对未扰动平均位置的高度， x 为沿传播方向的坐标，则沿 $+x$ 方向传播的无损正弦波可写为：

$$y(x,t) = A \cos \left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} + \phi_0 \right) \text{ (m),} \quad (1.17)$$

A 为振幅， T 为时间周期， λ 为波长， ϕ_0 为不随时间和空间变化的参考相位。也可写成式（1.18），其中相位 $\phi(x,t)$ 由式（1.19）给出：

$$y(x,t) = A \cos \phi(x,t) \quad \text{(m),} \quad (1.18) \quad \phi(x,t) = \left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} + \phi_0 \right) \quad \text{(rad).} \quad (1.19)$$

相位 $\phi(x,t)$ 随 x 、 t 变化，不应与常数 ϕ_0 混淆。角度可用弧度或度表示， $2\pi \text{ rad} = 360^\circ$ 。令 $\phi_0 = 0$ ，得到：

$$y(x,t) = A \cos \left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} \right) \quad \text{(m).} \quad (1.20)$$

图1-12分别给出 $t = 0$ 时随 x 变化的空间波形，以及 $x = 0$ 时随 t 变化的时间波形；两者的周期分别为 λ 、 T 。图1-13中的三个时刻虽然波峰位置不同，相邻波峰间距却都为 λ 。随着时间增大，波形向 $+x$ 移动，故称沿 $+x$ 传播的行波。

图1-12见左侧原书：（a）固定 $t = 0$ ，观察 $y(x,0)$ ；（b）固定 $x = 0$ ，观察 $y(0,t)$ 。横轴分别为位置 x 与时间 t ，纵轴均为位移。

跟踪同一波峰 P 或其他固定相位点，可求得相速度。波峰满足相位为 2π 的整数倍：

$$\phi(x,t) = \frac{2\pi t}{T} - \frac{2\pi x}{\lambda} = 2n\pi, \quad n = 0, 1, 2, \dots \quad (1.21)$$

同样，对任意选定高度 y_0 ，沿同一相位分支连续跟踪，便可把相位设为常数：

$$y(x,t) = y_0 = A \cos \left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} \right) \quad (1.22) \quad \frac{2\pi t}{T} - \frac{2\pi x}{\lambda} = \cos^{-1} \left(\frac{y_0}{A} \right) = \text{constant.} \quad (1.23)$$

沿该固定相位点的轨迹对时间求导，即得式（1.24）和相速度式（1.25）：

$$\frac{2\pi}{T} - \frac{2\pi}{\lambda} \frac{dx}{dt} = 0, \quad (1.24) \quad u_p = \frac{dx}{dt} = \frac{\lambda}{T} \quad \text{(m/s).} \quad (1.25)$$

相速度是波形或等相位点的传播速度。水质点主要在平衡位置附近振动，并不随波形整体向前迁移。

数学校注：一般波峰条件应取 n 为任意整数；固定 y_0 可对应多个相位值，推导时应固定其中一个连续分支，不能在不同波峰或不同反余弦分支间跳换。

relative to the mean height (undisturbed condition) and x denotes the distance of wave travel, the functional dependence of y on time t and the spatial coordinate x has the general form

$$y(x,t) = A \cos \left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} + \phi_0 \right) \text{ (m),} \quad (1.17)$$

where A is the **amplitude** of the wave, T is its **time period**, λ is its **spatial wavelength**, and ϕ_0 is a **reference phase**. The quantity $y(x,t)$ also can be expressed in the form

$$y(x,t) = A \cos \phi(x,t) \quad \text{(m),} \quad (1.18)$$

where

$$\phi(x,t) = \left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} + \phi_0 \right) \quad \text{(rad).} \quad (1.19)$$

The angle $\phi(x,t)$ is called the **phase** of the wave, and it should not be confused with the reference phase ϕ_0 , which is constant with respect to both time and space. Phase is measured by the same units as angles, that is, radians (rad) or degrees, with $2\pi \text{ radians} = 360^\circ$.

Let us first analyze the simple case when $\phi_0 = 0$:

$$y(x,t) = A \cos \left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} \right) \quad \text{(m).} \quad (1.20)$$

The plots in **Fig. 1-12** show the variation of $y(x,t)$ with x at $t = 0$ and with t at $x = 0$. The wave pattern repeats itself at a spatial period λ along x and at a temporal period T along t .

If we take time snapshots of the water's surface, the height profile $y(x,t)$ would exhibit the sinusoidal patterns shown in **Fig. 1-13**. All three profiles correspond to three different values of t , and the spacing between peaks is equal to the wavelength λ , even though the patterns are shifted relative to one another because they correspond to different observation times. Because the pattern advances along the $+x$ direction at progressively increasing values of t , $y(x,t)$ is called a wave traveling in the $+x$ direction. If we track a given point on the wave, such as the peak P , and follow it in time, we can measure the **phase velocity** of the wave. At the peaks of the wave pattern, the phase $\phi(x,t)$ is equal to zero or multiples of 2π radians. Thus,

$$\phi(x,t) = \frac{2\pi t}{T} - \frac{2\pi x}{\lambda} = 2n\pi, \quad n = 0, 1, 2, \dots \quad (1.21)$$

Had we chosen any other fixed height of the wave, say y_0 , and monitored its movement as a function of t and x , this again would have been equivalent to setting the phase $\phi(x,t)$ constant such that

$$y(x,t) = y_0 = A \cos \left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} \right) \quad (1.22)$$

or

$$\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} = \cos^{-1} \left(\frac{y_0}{A} \right) = \text{constant.} \quad (1.23)$$

The apparent velocity of that fixed height is obtained by taking the time derivative of Eq. (1.23), so

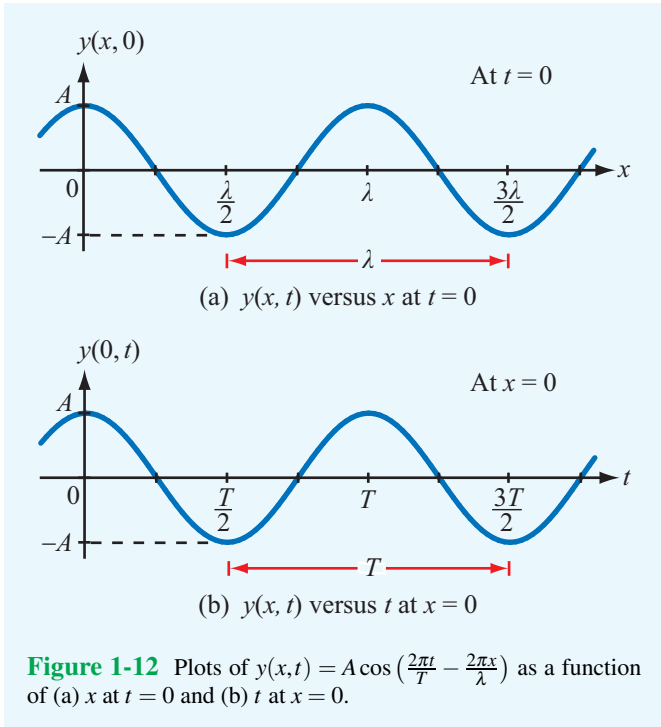
$$\frac{2\pi}{T} - \frac{2\pi}{\lambda} \frac{dx}{dt} = 0, \quad (1.24)$$

which gives the **phase velocity** u_p as

$$u_p = \frac{dx}{dt} = \frac{\lambda}{T} \quad \text{(m/s).} \quad (1.25)$$

► The phase velocity, also called the **propagation velocity**, is **the velocity of the wave pattern** as it moves across the water's surface. ◀

The water itself mostly moves up and down; when the wave moves from one point to another, the water does not move physically along with it.



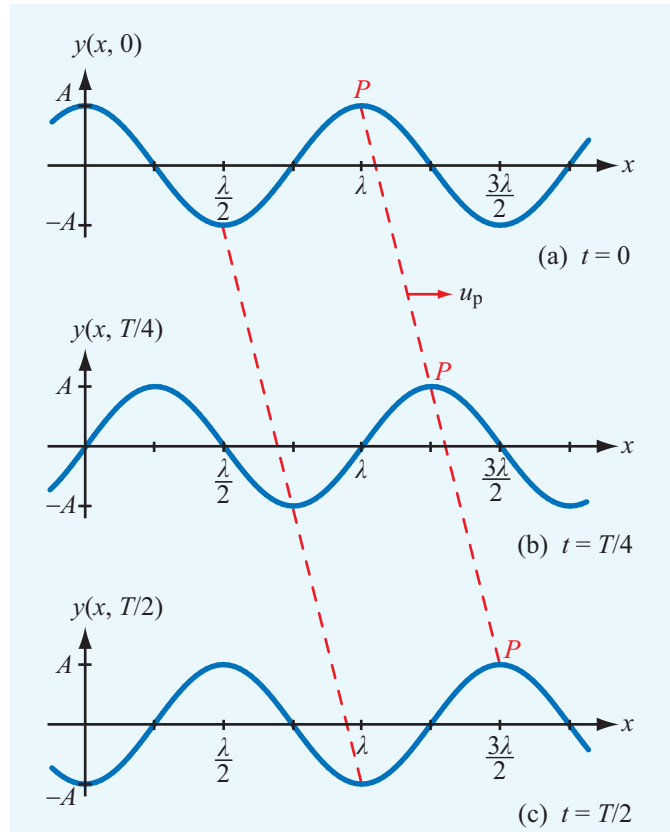


Figure 1-13 Plots of $y(x, t) = A \cos\left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda}\right)$ as a function of x at (a) $t = 0$, (b) $t = T/4$, and (c) $t = T/2$. Note that the wave moves in the $+x$ direction with a velocity $u_p = \lambda/T$.

The **frequency** of a sinusoidal wave, f , is the reciprocal of its time period T :

$$f = \frac{1}{T} \quad (\text{Hz}). \quad (1.26)$$

Combining the preceding two equations yields

$$u_p = f\lambda \quad (\text{m/s}). \quad (1.27)$$

The wave frequency f is measured in cycles per second and has been assigned the unit hertz (Hz), named in honor of the German physicist Heinrich Hertz (1857–1894), who pioneered the development of radio-wave instrumentation.

Using Eq. (1.26), Eq. (1.20) can be rewritten in a more

compact form as

$$\begin{aligned} y(x, t) &= A \cos\left(2\pi f t - \frac{2\pi}{\lambda} x\right) \\ &= A \cos(\omega t - \beta x), \end{aligned} \quad (1.28)$$

(wave moving along $+x$ direction)

where ω is the **angular velocity** of the wave and β is its **phase constant** (or **wavenumber**), defined as

$$\omega = 2\pi f \quad (\text{rad/s}), \quad (1.29a)$$

$$\beta = \frac{2\pi}{\lambda} \quad (\text{rad/m}). \quad (1.29b)$$

In terms of these two quantities,

$$u_p = f\lambda = \frac{\omega}{\beta}. \quad (1.30)$$

So far, we have examined the behavior of a wave traveling in the $+x$ direction. To describe a wave traveling in the $-x$ direction, we reverse the sign of x in Eq. (1.28):

$$\begin{aligned} y(x, t) &= A \cos(\omega t + \beta x). \end{aligned} \quad (1.31)$$

(wave moving along $-x$ direction)

► The direction of wave propagation is easily determined by inspecting the signs of the t and x terms in the expression for the phase $\phi(x, t)$ given by Eq. (1.19): If one of the signs is positive and the other is negative, then the wave is traveling in the positive x direction, and if both signs are positive or both are negative, then the wave is traveling in the negative x direction. The constant phase reference ϕ_0 has no influence on either the speed or the direction of wave propagation. ◀

We now examine the role of the phase reference ϕ_0 given previously in Eq. (1.17). If ϕ_0 is not zero, then Eq. (1.28) should be written as

$$y(x, t) = A \cos(\omega t - \beta x + \phi_0). \quad (1.32)$$

A plot of $y(x, t)$ as a function of x at a specified t or as a function of t at a specified x is shifted in space or time, respectively, relative to a plot with $\phi_0 = 0$ by an amount

频率、波数与传播方向

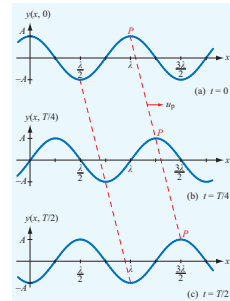


图1-13 $t = 0, T/4, T/2$ 时的波形；同一波峰P以 $u_p = \lambda/T$ 沿 $+x$ 方向移动。

正弦波频率 f 是周期 T 的倒数。结合相速度表达式可得：

$$f = \frac{1}{T} \quad (\text{Hz}). \quad (1.26) \quad u_p = f\lambda \quad (\text{m/s}). \quad (1.27)$$

频率表示每秒重复的周期数，单位为赫兹 (Hz)，得名于无线电波实验先驱 Heinrich Hertz (1857—1894)。利用式 (1.26)，式 (1.20) 可写为：

$$\begin{aligned} y(x, t) &= A \cos\left(2\pi f t - \frac{2\pi}{\lambda} x\right) \\ &= A \cos(\omega t - \beta x), \end{aligned} \quad (1.28)$$

(wave moving along $+x$ direction)

ω 称为角频率， β 称为相位常数或波数，分别定义为：

$$\omega = 2\pi f \quad (\text{rad/s}), \quad (1.29a)$$

$$\beta = \frac{2\pi}{\lambda} \quad (\text{rad/m}). \quad (1.29b)$$

用这两个量表示相速度，有：

$$u_p = f\lambda = \frac{\omega}{\beta}. \quad (1.30)$$

要描述沿 $-x$ 方向传播的波，只须将式 (1.28) 中 x 项的符号反转：

$$\begin{aligned} y(x, t) &= A \cos(\omega t + \beta x). \end{aligned} \quad (1.31)$$

(wave moving along $-x$ direction)

观察式 (1.19) 相位中 t 项和 x 项的符号即可判断传播方向：在 ω 、 β 取正值的约定下，两项异号表示沿 $+x$ 传播，同号表示沿 $-x$ 传播。常数参考相位 ϕ_0 不改变传播方向或相速度。

式 (1.17) 的参考相位 ϕ_0 不为零时，沿 $+x$ 方向传播的波为：

$$y(x, t) = A \cos(\omega t - \beta x + \phi_0). \quad (1.32)$$

固定时刻观察空间波形，或固定位置观察时间波形，相对于 $\phi_0 = 0$ 的参考波都会发生与 ϕ_0 成正比的平移。（接下页）